

Project Report: CodeQuest Grading Agent - Agentic AI Engineer

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Family:	Agentic AI Engineer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Agent reads student report submission, checks rubric criteria, drafts feedback - human approves before publish.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview** *(Level: Intermediate)

CodeQuest Grading Agent** is a intermediate-level end-to-end project for Agentic AI Engineer. **Duration: 3-4 weeks.** Agent reads student report submission, checks rubric criteria, drafts feedback - human approves before publish. **Tools: LangGraph, FastAPI, OpenAI, React.**

Code Reference:

```
Objective: Tools: load_report, check_rubric, draft_feedback
Objective: LangGraph state machine
Objective: Human-in-the-loop approval step
Objective: Audit log of all agent actions
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Agentic AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Agentic AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Tools [Intermediate]

Chapter focus: Phase: Tools** *(Level: Intermediate)

Report + rubric tools. Complete these tasks: JSON report loader; Rubric scorer. Deliverable for this phase: Tool unit tests.

Code Reference:

- [] JSON report loader
- [] Rubric scorer

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Tools is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Tool unit tests to the teacher.

Chapter 3: Phase: Graph [Intermediate]

Chapter focus: Phase: Graph** *(Level: Intermediate)

LangGraph workflow. Complete these tasks: Agent ? Tools ? Review node. Deliverable for this phase: Graph diagram.

Code Reference:

- [] Agent ? Tools ? Review node

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Graph is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Graph diagram to the teacher.

Chapter 4: Phase: HITL [Intermediate]**Chapter focus: Phase: HITL** *(Level: Intermediate)**

Approval gate. Complete these tasks: Teacher confirm UI; Reject/revise loop. Deliverable for this phase: Approval flow demo.

Code Reference:

- [] Teacher confirm UI
- [] Reject/revise loop

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase HITL is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Approval flow demo to the teacher.

Chapter 5: Phase: Audit [Intermediate]**Chapter focus: Phase: Audit** *(Level: Intermediate)**

Log every action. Complete these tasks: Tool calls; Prompts; Outputs. Deliverable for this phase: Audit log sample.

Code Reference:

- [] Tool calls
- [] Prompts
- [] Outputs

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Audit is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Audit log sample to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review** *(Level: Intermediate)

Submit all final deliverables: Grading agent; Graph diagram; Approval demo; Audit log. Skills demonstrated: LangGraph, HITL, Tool calling. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Grading agent
Deliverable: Graph diagram
Deliverable: Approval demo
Deliverable: Audit log
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Grading agent and passes the rubric.